Do bangs('!') get more clicks?

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Introduction

We want to answer the question: Are there differences in average clicks between headlines that use a "!" and headlines that do not contain a "!".

```
#tinytex::tlmgr_install("tabularray")
```

```
#install.packages("labelled")
#install.packages("tinytable")

library(haven)
library(labelled)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(tinytable)
```

```
upworthy <- read_csv("https://osf.io/vy8mj/download")
```

```
New names:
Rows: 105551 Columns: 17
-- Column specification
-------------------------------------------------------- Delimiter: "," chr
(8): clickability_test_id, excerpt, headline, lede, slug, eyecatcher_id... dbl
(5): ...1, impressions, clicks, significance, test_week lgl (2): first_place,
winner dttm (2): created_at, updated_at
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

```
upworthy |>
  names()
```

```
 [1] "...1"                "created_at"          "updated_at"
 [4] "clickability_test_id" "excerpt"             "headline"
 [7] "lede"                "slug"                "eyecatcher_id"
[10] "impressions"         "clicks"              "significance"
[13] "first_place"         "winner"              "share_text"
[16] "square"              "test_week"
```

Select variables that we need

- "created_at";
- "clickability_test_id", so that we can create comparison groups;
- "headline";
- "impressions", which is the number of people that saw the package; and
- "clicks" which is the number of clicks on that package.

```
upworthy_restricted <-
  upworthy |>
  select(
    created_at, clickability_test_id, headline, impressions, clicks
    )
head(upworthy_restricted)
```

```
# A tibble: 6 x 5
  created_at          clickability_test_id     headline       impressions clicks
  <dttm>              <chr>                    <chr>                <dbl>  <dbl>
1 2014-11-20 11:33:26 546dd17e26714c82cc00001c Let's See … H~        3118      8
2 2014-11-20 15:00:01 546e01d626714c6c4400004e People Sent T~        4587    130
```

```
3 2014-11-20 11:33:51 546dd17e26714c82cc00001c $3 Million Is~        3017     19
4 2014-11-20 11:34:12 546dd17e26714c82cc00001c The Fact That~        2974     26
5 2014-11-20 11:34:33 546dd17e26714c82cc00001c Reason #351 T~        3050     10
6 2014-11-20 11:34:48 546dd17e26714c82cc00001c I Was Already~        3061     20
```

We will focus on the text contained in headlines, and look at whether headlines contains "!" got more clicks than those that did not. We want to remove the effect of different images and so will focus on those tests that have the same image. To identify whether a headline contains a "!", we search for "!".

```
upworthy_restricted <-
  upworthy_restricted |>
  mutate(
    suprised =
      str_detect(string = headline, pattern = "\\!")
    )

upworthy_restricted |>
  count(suprised)
```

```
# A tibble: 2 x 2
  suprised      n
  <lgl>     <int>
1 FALSE    103514
2 TRUE       2037
```

For every test, and for every picture, we want to know whether "!" affected the number of clicks.

```
supprised_or_not <-
  upworthy_restricted |>
  summarise(
    ave_clicks = mean(clicks),
    .by = c(clickability_test_id, suprised)
  )

supprised_or_not |>
  pivot_wider(names_from = suprised,
              values_from = ave_clicks,
              names_prefix = "ave_clicks_") |>
  drop_na(ave_clicks_FALSE, ave_clicks_TRUE) |>
```

contains a '!' ?	Mean clicks
FALSE	57
TRUE	42

```
  mutate(difference_in_clicks = ave_clicks_TRUE - ave_clicks_FALSE) |>
  summarise(average_differce = mean(difference_in_clicks))
```

```
# A tibble: 1 x 1
  average_differce
             <dbl>
1            -2.93
```

```
supprised_or_not |>
  summarise(mean = mean(ave_clicks),
            .by = suprised) |>
  tt() |>
  style_tt(j = 1:2, align = "lr") |>
  format_tt(digits = 0, num_fmt = "decimal") |>
  setNames(c("contains a '!' ?", "Mean clicks"))
```